

CO250 Assignment 1

September 14–21

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- [1] Waterloo's infamous Maple Syrup Madness produces maple syrup for the entire southern part of Ontario. The following table shows the demand d_i (in tons) for each month $i \in \{1, \dots, 12\}$.

Month	1	2	3	4	5	6	7	8	9	10	11	12
d_i	10	12	14	20	15	18	19	20	30	28	31	22

The company needs to decide how much maple syrup to produce in each month so that the demand can be met. As usual we make the following assumption: maple syrup produced in month i can be consumed in month i (and later, of course).

- [a] Assume that producing a ton of maple syrup in month i costs p_i , and that the company can store up to 15 tons of its product in its storage facility for an unlimited amount of time. Assume that the storage facility contains 7 tons of maple syrup at the beginning of month 1.

The company wants to determine how much to produce in each month in order to minimize the total production cost. Write an LP for this task.

Answer:

Let s_i be the amount of maple syrup (in tons) in storage at the end of month i .

Let x_i be the amount of maple syrup produced in month i .

$$\begin{array}{ll} \text{Objective:} & \min \sum_{i=1}^{12} p_i x_i \\ \text{subject to:} & s_i = s_{i-1} + x_i - d_i \quad \forall i \in [12] \\ & 0 \leq s_i \leq 15 \quad \forall i \in [12] \\ & s_0 = 7 \\ & x_i \geq 0 \quad \forall i \in [12]. \end{array}$$

- [b] Now assume that the company wants to even out production over the year in order to keep its demand for staff constant. We model this by assuming that the company incurs a cost of 50 for each ton of difference between the production quantities in any two consecutive months. As an example, assume that the company decides to produce x_i tons of maple syrup in month $i \in \{1, \dots, 12\}$.

i	1	2	3	4	5	6	7	8	9	10	11	12
x_i	4	19	21	16	32	32	20	10	12	5	6	8

The cost incurred by the company for the production plan given in the above table is

$$\begin{aligned} & ((19 - 4) + (21 - 19) + (21 - 16) + (32 - 16) + (32 - 32) + (32 - 20) \\ & + (20 - 10) + (12 - 10) + (12 - 5) + (6 - 5) + (8 - 6)) \cdot 50. \end{aligned}$$

As in (a), assume that the storage facility contains 7 tons of maple syrup at the beginning of month 1, and that you can store up to 15 tons of maple syrup at any time.

The company wants to find a production schedule that minimizes the total cost incurred by production changes. Formulate the problem as an LP.

Answer:

Introduce auxiliary variables y_i that, through constraints, will represent $y_i = |x_i - x_{i+1}|$.

$$\begin{aligned} \text{Objective:} \quad & \min \sum_{i=1}^{11} 50y_i \\ \text{subject to:} \quad & y_i \geq x_i - x_{i+1} & \forall i \in [11] \\ & y_i \geq x_{i+1} - x_i & \forall i \in [11] \\ & s_i = s_{i-1} + x_i - d_i & \forall i \in [12] \\ & 0 \leq s_i \leq 15 & \forall i \in [12] \\ & s_0 = 7 \\ & x_i \geq 0 & \forall i \in [12]. \end{aligned}$$

- [2] Consider the following word game. You are assigned a number of tiles, each containing letters from the alphabet

$$\Sigma = \{a, b, c, \dots, z\}.$$

Suppose that your input contains $N_\alpha \in \mathbb{N}$ tiles with the letter α for each $\alpha \in \Sigma$. From the letters you are to construct any of the words $w_1, \dots, w_n$. This list might for example consist of all the words in a dictionary.

You must construct any word at most once, and use any tile at most once. You receive $v_j \geq 0$ points for constructing word w_j , and an additional bonus $b_{ij} \geq 0$ for making both words w_i and w_j .

- [a] Formulate an integer program to determine your optimal choice of words (one that maximizes the total number of points + bonus).

Answer:

Let

$$x_j = \begin{cases} 1 & \text{if we construct word } w_j \\ 0 & \text{else} \end{cases}$$

Let

$$y_{ij} = \begin{cases} 1 & \text{if we construct both words } w_i \text{ and } w_j \\ 0 & \text{else} \end{cases}$$

Let $a_{\alpha j}$ be the amount of times letter α appears in word w_j

$$\begin{aligned} \text{Objective:} \quad & \max \sum_{j \in [n]} v_j x_j + \sum_{1 \leq i < j \leq n} b_{ij} y_{ij} \\ \text{subject to:} \quad & y_{ij} \leq x_i \\ & y_{ij} \leq x_j \\ & y_{ij} \geq x_i + x_j - 1 \\ & \sum_{j \in [n]} a_{\alpha j} x_j \leq N_{\alpha} & \forall \alpha \in \Sigma \\ & x_j \in \{0, 1\} & \forall j \in [n] \\ & y_{ij} \in \{0, 1\} & \forall 1 \leq i < j \leq n \end{aligned}$$

- [b] How does your formulation change when you are allowed to select 100 tiles with no restriction on your choice of letters?

Answer:

$$\begin{aligned} \text{Objective:} \quad & \max \sum_{j \in [n]} v_j x_j + \sum_{1 \leq i < j \leq n} b_{ij} y_{ij} \\ \text{subject to:} \quad & y_{ij} \leq x_i \\ & y_{ij} \leq x_j \\ & y_{ij} \geq x_i + x_j - 1 \\ & \sum_{\alpha \in \Sigma} \sum_{j \in [n]} a_{\alpha j} x_j \leq 100 \\ & x_j \in \{0, 1\} & \forall j \in [n] \\ & y_{ij} \in \{0, 1\} & \forall 1 \leq i < j \leq n \end{aligned}$$

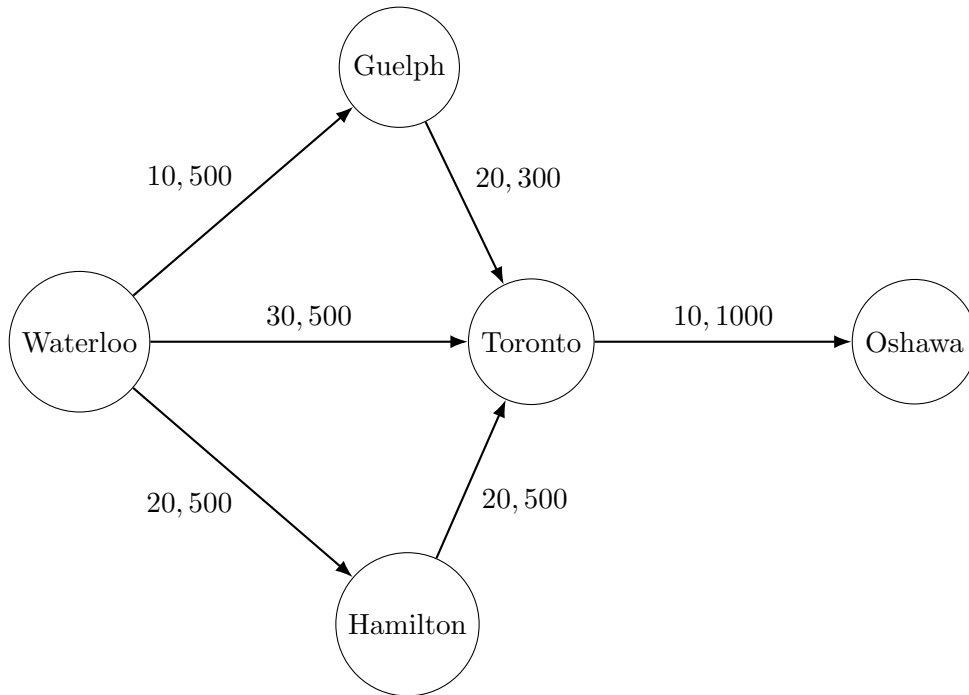


Figure 1: Maple Syrup Madness Pipe Network for Problem 3. The numbers 10, 500 on the arc connecting Waterloo to Guelph signal that it costs 10 cents to send a litre from Waterloo to Guelph, and that it is possible to send up to 500 litres.

[3] Maple Syrup Madness has a plant in each of the cities of Waterloo, Guelph, Hamilton, Toronto, and Oshawa. Maple Syrup Madness marketing team has proposed that selling maple syrup for 100 cents in Oshawa may be a good idea for the company brand. To attend this demand, the company will employ its remarkable pipe infrastructure connecting its plants (see Figure 1). Namely:

- the Waterloo plant can send maple syrup to Guelph, Hamilton, and Toronto, at the cost of 10, 20, and 30 cents per litre, respectively;
- the Guelph plant can send maple syrup to Toronto at the cost of 20 cents per litre;
- the Hamilton plant can send maple syrup to Toronto at the cost of 20 cents per litre;
- the Toronto plant can send maple syrup to Oshawa at the cost of 10 cents per litre.

The Waterloo plant has a total of 1000 litres of maple syrup that can be used in this publicity stunt. But given the short window between when these litres are produced and when they must be sold in Oshawa, the pipe capacities impose restrictions on how many litres can be sent in each pipe. Each of the pipes listed above has a maximum capacity of 500 litres; except the pipe between Guelph and Toronto, that can only be used to send 300 litres, and the recently upgraded pipe between Toronto and Oshawa, which can be used to send 1000 litres.

The company would like to know the maximum profit it can achieve from selling Waterloo's maple syrup in Oshawa for 100 cents. Formulate an LP to solve that task.

Answer:

Let v_{xy} represent the liters of maple syrup sent from x to y . So we have variables:

- v_{wg} = litres sent from Waterloo to Guelph
- v_{wt} = litres sent from Waterloo to Toronto
- v_{wh} = litres sent from Waterloo to Hamilton
- v_{gt} = litres sent from Guelph to Toronto
- v_{ht} = litres sent from Hamilton to Toronto
- v_{to} = litres sent from Toronto to Oshawa

Objective: $\max 100v_{to} - (10v_{wg} + 30v_{wt} + 20v_{wh} + 20v_{gt} + 20v_{ht} + 10v_{to})$

subject to: $\sum_{x \in \{g,t,h\}} v_{wx} \leq 1000$

$$v_{wg} \leq 500$$

$$v_{wt} \leq 500$$

$$v_{wh} \leq 500$$

$$v_{gt} \leq 300$$

$$v_{ht} \leq 500$$

$$v_{to} \leq 1000$$

$$v_{gt} = v_{wg}$$

$$v_{ht} = v_{wh}$$

$$v_{to} = v_{gt} + v_{wt} + v_{ht}$$

$$v_{xy} \geq 0$$

$\forall x, y$

[4] [a] Show that the vectors

$$A_1 := \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad A_2 := \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad A_3 := \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

form a basis for $\mathbb{R}^3$.

Answer:

[b] Express each of the following vectors as linear combinations of A_1 , A_2 , and A_3 :

$$A_4 := \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad A_5 := \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad A_6 := \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Answer:

[c] Let m and k be a pair of positive integers.

Let $\{A_1, A_2, \dots, A_k\} \subset \mathbb{R}^m$ be a linearly independent subset of vectors. Suppose $A_{k+1} \in \mathbb{R}^m$ is not in the subspace spanned by the vectors $\{A_1, A_2, \dots, A_k\}$. Prove that the set

$$\{A_1, A_2, \dots, A_k, A_{k+1}\}$$

is linearly independent.

Answer:

[d] Prove that every linearly independent subset of vectors in $\mathbb{R}^m$ is part of a basis. So, given a linearly independent subset of vectors (say k of them) in $\mathbb{R}^m$, we can always find $m - k$ vectors to extend the initial subset to a basis for $\mathbb{R}^m$.

Answer: